



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

17-04-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,D	A,B,C	C,D	A,B	A,B,C,D	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	A	D	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.00	1.00	1.00	4.00	4.00	5.82	1.50	2.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,D	A,B,C,D	B,D	A,B,C,D	A,B,C,D	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	C	C	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.40	1.80	181.14	6.67	300.00	138.50	6.10	4.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C	B	B,C	A,B,D	B,C,D	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	C	A	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.00	6.00	4320.00	1.00	7.00	0.00	2022.00	0.25

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,D)

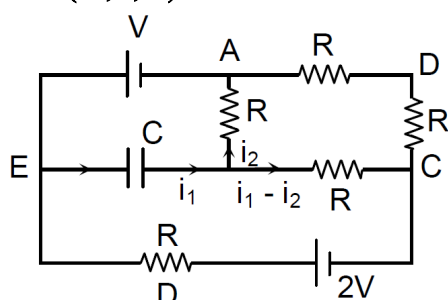
$$\frac{1000}{S_1} = \frac{2000}{S_2}, S_2 = 2S_1$$

$$S_1x + S_2y = (S_1 + S_2) 40$$

$$S_1x + 2S_1y = (S_1 + 2S_1) 40$$

$$x + 2y = 120$$

2. Ans (A,B,C)



$$\text{time constant of capacitor } (\tau) = \frac{5RC}{8}$$

Maximum charge on capacitor at steady state

$$(q_0) = 5CV / 4$$

Charge on capacitor at any time

$$q = \frac{5CV}{4} \left(1 - e^{-\frac{8t}{5RC}} \right) \dots\dots (i)$$

Current in the branch of capacitor at any time

$$i_1 = \frac{2V}{R} e^{-\frac{8t}{5RC}}$$

$$i_1 = 1.7 \text{ amp}$$

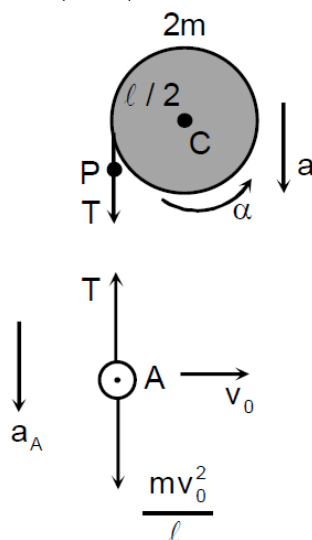
$$\text{Now, } A_A + i_2 R (i_1 - i_2) R = V_c$$

$$V_A - V_c = 2.52 \text{ volt}$$

3. **Ans (C,D)**

The force on the particle is not the central force.

4. **Ans (A,B)**



For the disc (2m)

$$T = 2ma \quad \dots(i)$$

$$T \times \frac{r}{2} = I_c \alpha \quad \dots(ii)$$

For A (w.r.t. 'P')

$$T + m[a + R\alpha] = \frac{mv_0^2}{r}$$

$$\text{Initial R.O.C of A} = \frac{V_0^2}{(T/m)}$$

5. **Ans (A,B,C,D)**

$$a_x + a_y = a_1 \quad \dots(i)$$

$$a_y - a_x = a_2 \quad \dots(ii)$$

$$a_{cm(x)} = 0$$

$$2a_2 = a_1 + a_x \quad \dots(iii)$$

6. **Ans (A,B)**

$$L = mv\theta r \quad \dots(i)$$

$$\frac{dL}{dt} = -\alpha v\theta r$$

$$\Rightarrow \frac{dL}{dt} = -\frac{\alpha L}{m} \text{ from (i)}$$

$$\Rightarrow \int_{L_0}^L \frac{dL}{L} = -\frac{\alpha}{m} \int_0^t dt$$

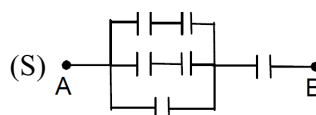
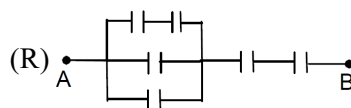
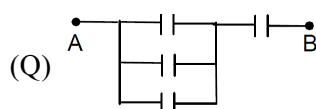
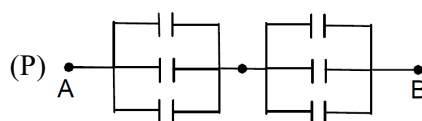
$$L = L_0 e^{-\frac{\alpha t}{m}}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (B)**

Eq. circuit of case P



8. **Ans (A)**

$$C = C_v + C_v + \frac{RT}{v} \frac{dv}{dT} = C_v + \frac{P dv}{dT} \text{ (for one}$$

mole gas)

$$\text{we have } p = P_0 + \frac{\alpha}{v}, \text{ or } \frac{RT}{v} = P_0 + \frac{\alpha}{v}$$

$$\text{or } RT = P_0 v + \alpha.$$

$$\text{Therefore, } R dT = P_0 dv, \text{ so, } \frac{dv}{dt} = \frac{R}{P_0}$$

Hence

$$C = C_v + \left(P_0 + \frac{\alpha}{v}\right) \frac{R}{P_0} = \frac{R}{r-1} + \left(1 + \frac{\alpha}{P_0 V}\right) R$$

(II) work done given by

$$W = \int_{v_1}^{v_2} \left(P_0 + \frac{\alpha}{v}\right) dv = P_0 (v_2 - v_1) + \alpha \ln \frac{v_2}{v_1}$$

$$(III) \Delta U = C_v (T_2 - T_1) =$$

$$C_v \left(\frac{P_2 v_2 - P_1 v_1}{R} \right) = \frac{1}{\gamma - 1} [p_2 v_2 - p_1 v_1]$$

$$= \frac{P_0 (v_2 - v_1)}{\gamma - 1}$$

$$(IV) Q = DV + W = \frac{\gamma P_0 (v_2 - v_1)}{\gamma - 1} + \alpha \ln \frac{v_2}{v_1}$$

9. Ans (D)

$$(A) (8 - 2t)\hat{j} - (8 - 2t)\hat{j}$$

$$(B) S = \frac{8 \times 8}{2 \times 2} = 16$$

$$(C) v = 8i - 8j, |v| = 8\sqrt{2}$$

$$(D) 0 = 8t - \frac{1}{2}2t^2, t = 8$$

10. Ans (B)

If energy required by transition or ionization is Q then kinetic energy K required for colliding.

Particle is

$$K = Q \left(1 + \frac{M_1}{M_2} \right)$$

here $M_1 = M_2$

$$\therefore K = 2Q$$

Minimum kinetic energy required for neutron to excite the atom = 2 × excitation energy.

Possible values of excitation energy = 10.2 eV,

12.1 eV,

Ionisation energy = 13.6 eV

PART-1 : PHYSICS

SECTION-II

1. Ans (2.00)

$$\frac{\delta F}{\delta x} = k_{\text{eff}} \text{ (numerically)}$$

Let us find δx (static deformation of the system).

Since the particle is in equilibrium,

$$\delta F = 2\delta F_s \sin \theta$$

By Pythagoras theorem, $x^2 + y^2 = \ell^2$

Taking differentials of both sides, we have

$$2x\delta x + 2y\delta y = 2\ell\delta\ell$$

Since y is constant, $\delta y = 0$

$$\delta\ell = \frac{x}{\ell}\delta x$$

$$\frac{\delta F}{\delta x} = 2k\sin^2\theta$$

$$k_{\text{eff}} = 2k\sin^2\theta$$

$$\text{Substituting } k_{\text{eff}}, \text{ we have } T = \frac{\pi}{\sin\theta} \sqrt{\frac{2m}{k}}$$

2. Ans (1.00)

$$\text{Velocity} = \omega\sqrt{A^2 - x^2}$$

$$= \frac{2\sqrt{3}\pi A}{T} \text{ (doubles)}$$

$$\frac{2\sqrt{3}\pi A}{T} = \frac{2\pi}{T} \sqrt{A^2 - x^2}$$

$$A_1 = \frac{\sqrt{13}}{2} A$$

$$\cos\theta = \frac{A/2}{A_1} = \frac{1}{\sqrt{13}} \quad \left(\theta = \cos^{-1} \left(\frac{1}{\sqrt{13}} \right) \right)$$

$$\cos^{-1} \left(\frac{1}{\sqrt{13}} \right) = \omega t$$

$$t = T \cos^{-1} \left(\frac{1}{\sqrt{13}} \right)$$

3. Ans (1.00)

According to given condition,

$$\frac{i \times 4}{i \times R_0} = \frac{AD}{DC} = \frac{40}{60} = \frac{2}{3}$$

$$\Rightarrow R = 12$$

$$\text{Now, } R = R_0 (1 + \alpha \Delta T)$$

$$12 = 6 (1 + \alpha 100)$$

$$\Rightarrow \alpha = 1 \times 10^{-2}$$

$$\text{So, } x = 1$$

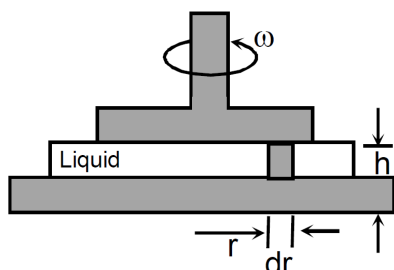
4. Ans (4.00)

Consider an element of disc at a radius r and having a width dr .

Linearly velocity at this radius = ωr

Shear stress, $\tau = \mu \frac{du}{dy}$

Assuming, the gap h to be small, so that the velocity distribution may be assumed linear



$$\tau = \mu \times \frac{v}{h} = \mu \frac{\omega r}{h}$$

Viscous force, $dF = \tau \times \text{Area}$

$$= \tau \times 2\pi r dr$$

Torque dT on the element,

$$dT = dF \times r = \tau 2\pi r^2 dr$$

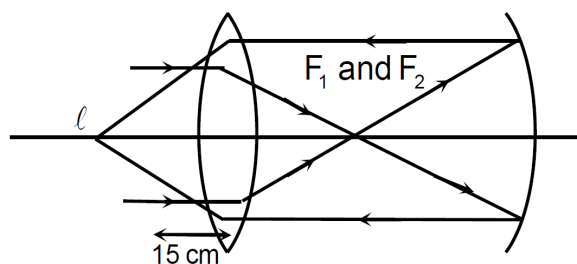
$$\text{or } dT = \frac{\mu \omega r}{h} \times 2\pi r^2 dr = \frac{2\pi \mu \omega^2 dr}{h}$$

Total torque

$$T = \int_0^{d/2} \frac{2\pi \mu \omega^2 dr}{h} = \frac{\mu \pi d^4 \omega}{4h}$$

Thus, $x = 4$

5. Ans (4.00)



The condition is possible only when the foci of lens and mirror coincides. Then, only the final image will be at 15 cm left to lens

$$\text{So, } x = 15 + 21 = 36 \text{ cm}$$

$$\text{Hence, } \frac{x}{9} = \frac{36}{9} = 4 \text{ cm}$$

6. Ans (5.82)

$$y_{\text{net}} = y_1 + y_2 + y_3$$

$$= a[\sin \omega t + \sin(\omega t + 45^\circ) + \sin(\omega t + 90^\circ)]$$

$$= (\sqrt{2} + 1) a \sin(\omega t + 45^\circ)$$

Energy ratio is

$$\frac{E_{\text{resultant}}}{E_{\text{ring}}} = \left(\frac{A}{a} \right)^2 = (\sqrt{2} + 1)^2 = (3 + 2\sqrt{2})$$

$$\therefore E_{\text{resultant}} = 5.82 \text{ J}$$

7. Ans (1.50)

Thermal stress = $Y \alpha \Delta \theta$

As, thermal stresses are equal

$$y_1 \alpha_1 \Delta \theta_1 = y_2 \alpha_2 \Delta \theta_2$$

$$\Rightarrow \frac{Y_1}{Y_2} = \frac{\alpha_2}{\alpha_1} = \frac{3}{2}$$

8. Ans (2.00)

Dimension of $B = [M^0 L^0 T^0]$

Dimension of $e = [AT]$

Dimension of $\epsilon_0 = [A^2 M^{-1} L^{-3} T^4]$

Dimension of $h = [ML^2 T^{-1}]$

Dimension of $c = [LT^{-1}]$

Equating,

$$[M^0 L^0 T^0] = \frac{[AT]^n}{[A^2 M^{-1} L^{-3} T^4][ML^2 T^{-1}][LT^{-1}]}$$

$$\Rightarrow [M^0 L^0 T^0] = [A^{n-2} L^0 T^{n-2}]$$

$$\therefore n - 2 = 0 \text{ or } n = 2$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B,D)

Isotones have same number of neutrons

2. Ans (A,B,C,D)

The $d_{A-A} = d_{B-B} = \sqrt{2} a/2$ and $d_{AB} = a/2$. There will be four molecules of AB in a unit cell.

$$\text{Hence, Its density will be given as } \rho = \frac{4 \times M_{Ab}}{a^3 N_A}.$$

3. **Ans (B,D)**

RLVP = mole fraction of solute

Van't Hoff factor for CH_3COOH in benzene will be less than 1, because of association of CH_3COOH

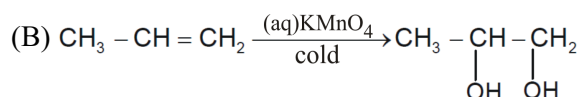
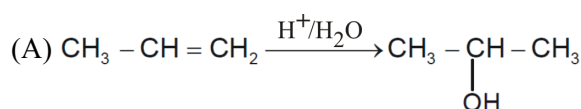
4. **Ans (A,B,C,D)**

Factual

5. **Ans (A,B,C,D)**

Factual

6. **Ans (B,D)**



PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (C)**

Factual

8. **Ans (C)**

Factual

9. **Ans (C)**

Factual

10. **Ans (C)**

Factual

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (3.40)**

$$\text{I.P} = R_H Z^2, 13.6 = R_H$$

$$\Delta E = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = 13.6 \left(\frac{1}{2^2} - \frac{1}{\infty^2} \right) = 13.6 \times \frac{1}{4} = 3.4$$

2. **Ans (1.80)**

$$\rho = \frac{PM}{RT} \text{ or } M = \frac{\rho \times RT}{P} = \frac{1.5 \times R \times 273}{1}$$

$$\rho = \frac{1000 \times 1.5 \times R \times 273}{760 \times R \times 300} = 1.8$$

3. **Ans (181.14)**

$$\text{for bcc, } \sqrt{3} \times a = 2(r_+ + r_-)$$

$$1.732 \times 387 = 2(154 + r_{\text{Cl}^-})$$

$$r_{\text{Cl}^-} \approx 181.14$$

4. **Ans (6.67)**

$$P_{\text{PCl}_5} = P_{\text{Cl}_3} = P_{\text{Cl}_2} = \frac{2}{6} \times 3 = 1 \text{ atm}$$

$$K_p = \frac{1 \times 1}{1} = 1$$

Let "y" mol of Cl_2 were added

	$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$	
initial mol	2	2 (2 + y)
mole of initial	(2 + x)	(2 - x) (2 + y - x)

Total mol = 12, because total volume is twice of initial volume

$$\text{Mol of } \text{PCl}_5 + \text{PCl}_3 = 2 + x + 2 - x = 4$$

$$\text{Mol of } \text{Cl}_2 = 12 - 4 = 8$$

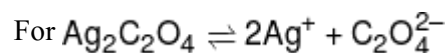
$$P_{\text{PCl}_5} = \frac{(2+x) \times 3}{12} = \frac{2+x}{4}, P_{\text{PCl}_3} = \frac{2-x}{4}, P_{\text{Cl}_2} = \frac{8 \times 3}{12} = 2$$

$$\frac{\left(\frac{2-x}{4} \right) \times 2}{\frac{2+x}{4}} = 1.0 \Rightarrow x = \frac{2}{3}$$

$$2 + y + \frac{2}{3} = 8$$

$$y = \frac{20}{3} = 6.67$$

5. **Ans (300.00)**



$$S = \left(\frac{K_{\text{sp}}}{4} \right)^{\frac{1}{3}} = \left(\frac{1.35 \times 10^{-11}}{4} \right)^{\frac{1}{3}} = (3.375 \times 10^{-12})^{\frac{1}{3}} = 1.5 \times 10^{-4}$$

$$\Lambda_m = \frac{k \times 1000}{S}$$

$$\Lambda_m = \frac{4.5 \times 10^{-5} \times 10^3}{1.5 \times 10^{-4}} = 300 \text{ S cm}^2\text{mol}^{-1}$$

6. Ans (A,B)

Since, $\vec{a}$ and $\vec{b}$ are unit vectors, we have

$$|\vec{a} - \vec{b}| = \sqrt{(\vec{a} - \vec{b})^2}$$

$$\therefore \sqrt{(\vec{a})^2 + (\vec{b})^2 - 2\vec{a} \cdot \vec{b}} = \sqrt{1 + 1 - 2 \cos 2\theta} = 2 |\sin \theta|$$

$$\text{Therefore, } |\vec{a} - \vec{b}| < 1$$

$$\Rightarrow 2 |\sin \theta| < 1$$

$$|\sin \theta| < \frac{1}{2}$$

$$\Rightarrow \theta \in \left[0, \frac{\pi}{6}\right) \quad \text{or} \quad \left(\frac{5\pi}{6}, \pi\right]$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

I.

$$OA = 1 + 4 \cot \theta$$

$$OB = 4 + \tan \theta$$

$$OA + OB = 5 + 4 \cos \theta + \tan \theta \geq 5 + 2\sqrt{4 \cot \theta \tan \theta} \\ = 5 + (2 \times 2) = 9$$

II.

The reflection of P(4, -1) on $y = x$ is Q(-1, 4).

$$\text{Hence, } PQ = \sqrt{(4+1)^2 + (-1-4)^2} = \sqrt{50} = 5\sqrt{2}.$$

III.

$$AB = 2\sqrt{2}$$

$$OC = \sqrt{2}$$

The maximum value of d is

$$OF = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$$

IV.

$$\text{The given line is } x = 4 + \frac{1}{\sqrt{2}} \left(\frac{y+1}{\sqrt{2}} \right) \text{ or } y = 2x - 9$$

Hence, the intercept made by the x-axis is 9/2.

8. Ans (C)

I. We have

$$\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi} = \log_\pi 3 + \log_\pi 4 = \log_\pi 12$$

But $\pi^2 < 12 < \pi^3$, we have $2 < \log_\pi 12 < 3$.

$$\text{II. } 3^a = 4; a = \log_3 4$$

Similarly, $b = \log_4 5$ etc.

$$\text{Hence, } abcdef = \log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot$$

$$\log_7 8 \cdot \log_8 9 = \log_3 9 = 2$$

III. We have to find characteristic of $\log_2 2008$.

We know that $\log_2 1024 = 10$ and $\log_2 2048 = 11$,

therefore $10 < \log_2 2008 < 11$

Hence, it has characteristic 10.

IV.

$$\log_2 (\log_2 (\log_3 x)) = 0 \Rightarrow \log_2 (\log_3 x) = 1$$

$$\Rightarrow \log_3 x = 2$$

$$\Rightarrow x = 9$$

Similarly, we have $\log_3 (\log_2 y) = 1$

$$\Rightarrow \log_2 y = 3 \text{ or } y = 8$$

Therefore, $x - y = 1$.

9. Ans (A)

$$\begin{aligned} \text{I. } 1 &= \lim_{x \rightarrow 0} \frac{e^{x^2} - 1 + x - e^x + 1}{2 \frac{\sin^2 x}{x^2} \cdot x^2} \\ &= \frac{1}{2} \left[\lim_{x \rightarrow 0} \frac{e^{x^2} - 1}{x^2} + \lim_{x \rightarrow 0} \frac{x - e^x + 1}{x^2} \right] \\ &= \frac{1}{2} \left[1 - \lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2} \right] \end{aligned}$$

$$\begin{aligned} \text{II. } 1 &= \lim_{x \rightarrow 0} \left(\frac{3+x}{3-x} \right)^{1/x} = e^{\lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{3+x}{3-x} - 1 \right)} \\ &= e^{\lim_{x \rightarrow 0} \frac{2x}{(3-x)^2}} = e^{2/3} \Rightarrow 2 + 3 = 5 \end{aligned}$$

$$\begin{aligned} \text{III. } \lim_{x \rightarrow 0} \frac{(\tan^3 x - x^3) - (\tan x^3 - x^3)}{x^5} \\ &= \lim_{x \rightarrow 0} \frac{\tan^3 x - x^3}{x^5} - \underbrace{\lim_{x \rightarrow 0} \frac{\tan^3 x - x^3}{x^5}}_{\text{zero (by expansion)}} \\ &= \lim_{x \rightarrow 0} \frac{(\tan x - x)}{x^3} \cdot \frac{(\tan^2 x + x \tan x + x^2)}{x^2} = \frac{1}{3} \times 3 = 1 \end{aligned}$$

IV. Rationalising given

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(x + 2 \sin x) \left[\sqrt{(x^2 + 2 \sin x + 1)} + \sqrt{\sin^2 x - x + 1} \right]}{(x^2 + 2 \sin x + 1) - (\sin^2 x - x + 1)} \\ &= 2 \cdot \lim_{x \rightarrow 0} \frac{x + \sin 2x}{x^2 - \sin^2 x + 2 \sin x + x} \\ &= 2 \cdot \lim_{x \rightarrow 0} \frac{1 + \frac{\sin 2x}{x}}{x - \frac{\sin^2 x}{x} + 2 + 1} = 2 \left(\frac{1+2}{3} \right) = 2 \end{aligned}$$

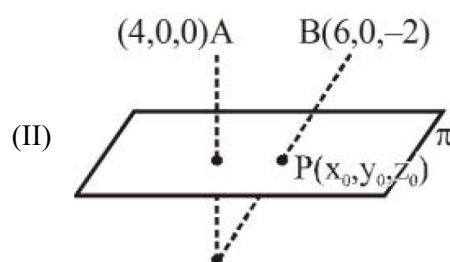
10. Ans (C)

Let the plane be $\alpha x + \beta z + 1 = 0$

Pass through $(1, 0, 1); (3, 2, -1)$

$$\therefore \alpha = -\frac{1}{2}; \beta = -\frac{1}{2}$$

$$\text{(I) } \pi : x + z = 2$$



Both A and B are on same side of π .

Reflection of A in plane is $A'(2, 0, -2)$

$$\text{Equation of line } A'B : \vec{r} = 6\hat{i} - 2\hat{k} + \lambda(4\hat{i})$$

$$\text{For P : } 6 + 4\lambda + 0 - 2 = 2 \Rightarrow \lambda = \frac{-1}{2}$$

$$\therefore P(4, 2, -2)$$

$$\therefore |4x_0 + y_0 + 2z_0| = 12$$

$$\text{(III) Now } |PA - PB|_{\min} = 0$$

$|PA - PB|_{\min}$ will approach

$$AB = \sqrt{4 + 0 + 4} = \sqrt{8}$$

$$\therefore |PA - PB| \in [0, \sqrt{8})$$

(IV) Also, A' will lie on

$$\begin{aligned} \frac{x-2}{1} &= \frac{y-\alpha}{0} = \frac{z+\beta}{-1} \\ \Rightarrow \frac{2-2}{1} &= \frac{0-\alpha}{0} = -\frac{2+\beta}{-1} \end{aligned}$$

$$\Rightarrow \alpha = 0, \beta = 2$$

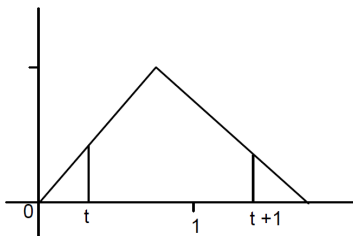
$$\therefore \alpha^4 + \beta^4 = 16$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (2.00)**

The region M is the large triangle in the diagram below, and the region N is the infinite strip between the lines $x = t$ and $x = t + 1$ where $0 < t < 1$



Hence $M \cap N$ is the pentagon. Its area is given by

$$\frac{1}{2} \left(\sqrt{2}^2 - t^2 - (2 - (t + 1))^2 \right) = -t^2 + t + \frac{1}{2}$$

2. **Ans (6.00)**

$$t^2 + 3t - 8 = 2 \Rightarrow t = 2, -5$$

$$2t^2 - 2t - 5 = -1 \Rightarrow t = 2, -1$$

So, at $t = 2$

$$\frac{dy}{dx} = \left(\frac{\frac{dy}{dt}}{\frac{dx}{dt}} \right)_{t=2} = \frac{6}{7}$$

3. **Ans (4320.00)**

a_3 is coefficient of x^3

$$\frac{10!}{8!} \times 3 \times 4 + \frac{10!}{7!3!} \times 3^3 = 4320$$

4. **Ans (1.00)**

Applying $C_1 \rightarrow C_1 - \sin \theta C_3$ and

$C_2 \rightarrow C_2 + \cos \theta C_3$, we get

$$f(\theta) = \begin{vmatrix} 1 & 0 & -\sin \theta \\ 0 & 1 & \cos \theta \\ \sin \theta & -\cos \theta & 0 \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 - \sin \theta R_1 + \cos \theta R_2$, we get

$$f(\theta) = \begin{vmatrix} 1 & 0 & -\sin \theta \\ 0 & 1 & \cos \theta \\ \sin \theta & -\cos \theta & \sin^2 \theta + \cos^2 \theta \end{vmatrix} = 1$$

$$\text{Thus, } f\left(\frac{\pi}{6}\right) = 1$$

5. **Ans (7.00)**

We have

$$(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$\Rightarrow 16 = 6 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$\therefore \alpha\beta + \beta\gamma + \gamma\alpha = 5$$

$$\text{Also, } \alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma = (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \gamma\alpha)$$

$$\Rightarrow 8 - 3\alpha\beta\gamma = 4(6 - 5)$$

$$\text{or } 3\alpha\beta\gamma = 4$$

$$\text{or } \alpha\beta\gamma = 4/3$$

$$\text{Now, } (\alpha^2 + \beta^2 + \gamma^2)^2 = \sum \alpha^4 + 2\sum \beta^2 \gamma^2$$

$$= \sum \alpha^4 + 2 \left\{ (\sum \beta\gamma)^2 - 2\alpha\beta\gamma \sum \alpha \right\}$$

$$(6)^2 = \sum \alpha^4 + 2 \left\{ 25 - 2 \cdot \frac{4}{3} \cdot 4 \right\}$$

$$\sum \alpha^4 = 36 - 50 + \frac{64}{3} = \frac{64}{3} - 14 = \frac{22}{3}$$

$$\therefore [\alpha^4 + \beta^4 + \gamma^4] = \left[\frac{22}{3} \right] = 7$$

6. **Ans (0.00)**

$$\text{Coefficient of } x^{103} \text{ in } (1 + x + x^2 + x^3 + x^4)^{199} (x - 1)^{201}$$

$$\text{Coefficient of } x^{103} \text{ in } -(1 - x)^2 (1 - x^5)^{199}$$

$$\text{Coefficient of } x^{103} \text{ in } -(1 - 2x + x^2) (1 - {}^{199}C_1 x^5 +$$

$${}^{199}C_2 x^{10} - {}^{199}C_3 x^{15} + \dots)$$

$$\text{Coefficient of } x^{103} = 0$$

7. **Ans (2022.00)**

For $x > 0$ not possible

For $x < 0$ put $x = -y$

$$\left(1 - \frac{1}{y}\right)^{1-y} = \left(\frac{y-1}{y}\right)^{1-y} = \left(\frac{y}{y-1}\right)^{1-y} = \left(\frac{2022}{2021}\right)^{2021}$$

$$\Rightarrow y = 2022$$

$$x = -2022$$

$$|x| = 2022$$

8. **Ans (0.25)**

$$P(B|A \cup \bar{B}) = \frac{P(B \cap (A \cup \bar{B}))}{P(A \cup \bar{B})} = \frac{P(A) - P(A \cap \bar{B})}{P(A) + P(\bar{B}) - P(A \cap \bar{B})}$$

$$= \frac{0.7 - 0.5}{0.7 + 0.6 - 0.5} = \frac{0.2}{0.8} = \frac{1}{4}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	D	A	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,C,D	A,C	A,D	A,C	A,C	A,B,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	5	4	1	2	2	4	4	6

PART-2 : CHEMISTRY

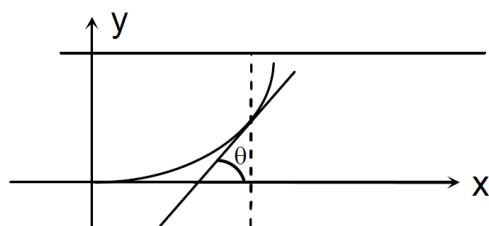
SECTION-I (i)	Q.	1	2	3	4				
	A.	D	A	B	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	C,D	A,C,D	A,D	A,D	A,C	A,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	7	6	6	5	6	4	4	3

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	B	B	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B	A,B,C,D	A,C	A,C	A,B,C,D	B,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	9	2	2	9	0	8	2	1

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (D)



Equation of circle is

$$(x - 0)^2 + (y - R)^2 = R^2 \quad \dots (1)$$

$$\frac{dy}{dx} = \tan \theta \quad \dots (2)$$

$$\text{by Snell's law } \sin(90 - \theta) \times \eta = 1 \times \eta_0 \quad \dots (3)$$

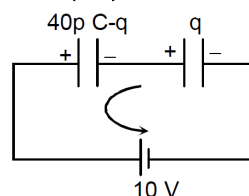
2. Ans (D)

$$F = - \left(\frac{Ay}{L} \right) \Delta L = K \Delta L$$

$$\Rightarrow K = \frac{Ay}{L}$$

$$T = 2\pi \sqrt{\frac{m}{K}}$$

3. Ans (A)



Apply KVL

$$\frac{40 - q}{2} - 10 - \frac{q}{2} = 0$$

Solve for q

4. Ans (B)

$$\text{Velocity, } v = \sqrt{\frac{YRT}{M}}$$

Find Y

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,C,D)

$$U = U_0(1 - \cos ax), F = \frac{-dU}{dx} = -aU_0 \sin ax$$

$$m \frac{d^2x}{dt^2} = -a^2 U_0 x \quad (\text{For small oscillations})$$

$$\Rightarrow T = 2\pi \sqrt{\frac{m}{a^2 U_0}}$$

Potential energy is maximum at $x = \frac{\pi}{a}$

amplitude is $\frac{\pi}{a}$

Speed of particles is maximum at $x = 0$

($\because$ PE is minimum)

6. Ans (A,C)

(i) for $\theta = 30^\circ$

$$\lambda = \frac{c}{v} = \frac{3 \times 10^8}{10^6} = 300 \text{ m, } d = 150 \text{ m}$$

$$\therefore \delta = \frac{2\pi}{\lambda}(d \sin \theta) = \frac{2\pi}{(300)}(150) \left(\frac{1}{2}\right) = \frac{\pi}{2}$$

$$I(\theta) = I_0 \cos^2 \left(\frac{\delta}{2}\right) = I_0 \cos^2 \left(\frac{\pi}{4}\right) = \frac{I_0}{2}$$

$$(ii) \text{ for } \theta = 90^\circ \Rightarrow \delta \left(\frac{2\pi}{300}\right)(150)(1) = \pi$$

$$\therefore I(\theta) = I_0 \cos^2 \left(\frac{\delta}{2}\right)$$

$$= I_0 \cos^2 \left(\frac{\pi}{2}\right) = 0$$

(iii) for $\theta = 0^\circ$, $\delta = 0$

$$\therefore I(\theta) = I_0$$

7. Ans (A,D)

Change in angular momentum = angular impulse

$$L = 5\tau$$

$$I = \frac{ma^2}{3}$$

$$K = \frac{L^2}{2I} = \frac{75\tau^2}{2ma^2}$$

8. Ans (A,C)

$$a = \frac{F_0}{m} \cos \omega t, \quad \int_0^{t_0} dv = \frac{F_0}{m} \int_0^{t_0} \cos \omega t dt,$$

$$t_0 = \frac{\pi}{\omega}$$

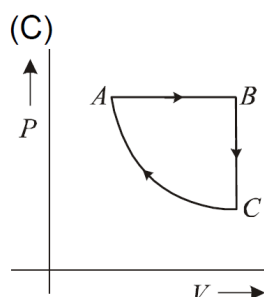
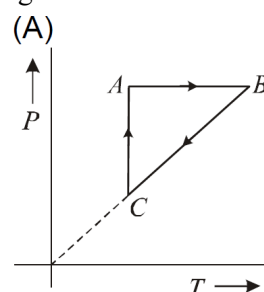
$$\text{and } \int_0^v dv = \frac{F_0}{m} \int_0^1 \cos \omega t dt, \quad v = \frac{F_0}{m\omega} \sin \omega t,$$

$$\int_0^s ds = \frac{F_0}{m\omega} \int_0^{t_0} \sin \omega t dt$$

$$S = \frac{F_0}{m\omega^2} [-\cos \omega t]_0^{t_0} = \frac{F_0}{m\omega^2} [2] = \frac{2F_0}{m\omega^2}$$

9. Ans (A,C)

Using $PV = nRT$



10. Ans (A,B,C,D)

Charge on capacitor at time t is

$$q = Q_0 \cos \omega t$$

$$\text{Where } \omega = \frac{1}{\sqrt{LC}}$$

$$\text{At } \omega t = \frac{\pi}{2}, q = 0$$

$$\Rightarrow t = \frac{\pi}{2} \sqrt{LC}$$

By conservation of energy

$$\frac{Q_0^2}{2C} = \frac{1}{2} Li_{\max}^2$$

$$\Rightarrow i_{\max} = \frac{Q_0}{\sqrt{LC}}$$

PART-1 : PHYSICS

SECTION-III

1. Ans (5)

$$(\Delta Q)_P = nC_p \Delta T$$

When piston is clamped; heat is rejected and volume becomes constant

$$(\Delta Q)_V = nC_v \Delta T$$

$$\text{Difference} = nC_p \Delta T - nC_v \Delta T = \Delta W = P \Delta V$$

$$P = P_0 + \frac{Mg}{A} \text{ and } \Delta V = (A \times 0.1)$$

$$\Rightarrow (\Delta Q)_P - (\Delta Q)_V = \left(P_0 + \frac{Mg}{A} \right) \times A \times 0.1$$

$$= 55 \text{ J}$$

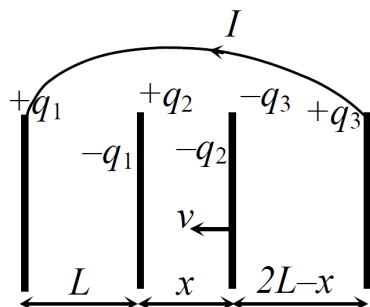
$$\Rightarrow x = 5$$

2. Ans (4)

$$\sqrt{2} \lambda_0 T_1 = \lambda_0 T_2$$

$$\frac{T_2}{T_1} = \sqrt{2} \quad ; \quad \frac{A_2}{A_1} = \left(\frac{T_2}{T_1} \right)^4 = 4$$

3. Ans (1)



$$\frac{2q_1 L}{2A\epsilon_0} + \frac{2q_2 x}{2A\epsilon_0} - \frac{2q_2 (2L - x)}{2A\epsilon_0} = 0 \quad \dots(i)$$

$$I = \frac{dq_1}{dt} = \frac{-dq_3}{dt} = \frac{dq_2}{dt} \quad \dots(ii)$$

$$\text{From (i), } q_1 L + q_2 x - 2q_3 L + q_3 x = 0$$

$$\Rightarrow L \frac{dq_1}{dt} + x \frac{dq_2}{dt} + q_2 \frac{dx}{dt} - 2 \frac{dq_3}{dt} L + x \frac{dq_3}{dt} + q_3 \frac{dx}{dt} = 0$$

$$\Rightarrow I = \frac{v}{3L} (q_2 + q_3) = 1 \text{ Ma}$$

4. Ans (2)

Apply Bernoulli's equations at pipe (1), (2) and (3)

$$p_3 + \frac{1}{2} \rho v_3^2 = p_2 + \frac{1}{2} \rho v_2^2 = p_1 + \rho l 2 v_1^2$$

$$p_0 + \rho gh_3 + \frac{1}{2} \rho v_3^2 = p_0 + \rho gh_2 + \frac{\rho}{2} v_2^2 = p_0 + \rho gh_1 + \frac{\rho}{2} v_1^2$$

$$gh_3 + \frac{v_3^2}{2} = gh_2 + \frac{v_2^2}{2} = gh_1 + \frac{v_1^2}{2} \quad \dots(i)$$

$$v_3 A_3 = v_2 A_2$$

$$v_3 A = v_2 \times 2 \quad \dots(ii)$$

$$v_2 \times 2 = v_1 \times 1 \quad \dots(iii)$$

From Equation (i), we get

$$gh_2 + \frac{v_2^2}{2} = gh_1 + \frac{v_1^2}{2}$$

$$10 \times \frac{35}{100} + \frac{v_2^2}{2} = 10 \times \frac{20}{100} + \frac{v_1^2}{2}$$

$$\frac{15}{10} = \frac{v_1^2 - v_2^2}{2}$$

$$= \frac{4v_2^2 - v_2^2}{2}$$

$$= \frac{3v_2^2}{2}$$

$$v_2 = 1 \text{ m/s}, v_1 = 2 \text{ m/s}$$

From equation (i), we get

$$gh_3 + \frac{v_3^2}{2} = gh_2 + \frac{v_2^2}{2}$$

$$= 10 \times \frac{37.5}{100} + \frac{v_3^2}{2}$$

$$= 10 \times \frac{35}{100} + \frac{1}{2}$$

$$\frac{v_3^2}{2} = \frac{1}{2} + \frac{35}{10} - \frac{37.5}{10} = \frac{1}{2} - \frac{2.5}{10} = \frac{1}{2} - \frac{25}{100}$$

$$\frac{v_3^2}{2} = \frac{1}{4} \Rightarrow v_3^2 = \frac{1}{2}$$

$$\Rightarrow v_3 = \frac{1}{\sqrt{2}}, X = 2$$

5. Ans (2)

$$\sqrt{v} = a(Z - b)$$

$$\sqrt{v} = a(Z - 1)$$

$$\Rightarrow (Z - 1)^2 \propto v$$

$$\text{or } (Z - 1)^2 \propto \frac{1}{\lambda}$$

$$\Rightarrow \frac{(Z_{\text{Mo}} - 1)^2}{(Z_{\text{Cu}} - 1)^2} = \frac{\lambda_{\text{Cu}}}{\lambda_{\text{Mo}}}$$

$$\Rightarrow \lambda_{\text{Cu}} = \lambda_{\text{Mo}} \frac{(Z_{\text{Mo}} - 1)^2}{(Z_{\text{Cu}} - 1)^2}$$

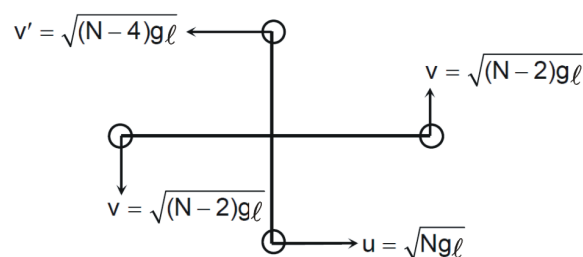
$$= 0.71 \times \left(\frac{41}{28}\right)^2 = 1.52 \text{ \AA}$$

$$\lambda = 2 \text{ \AA}$$

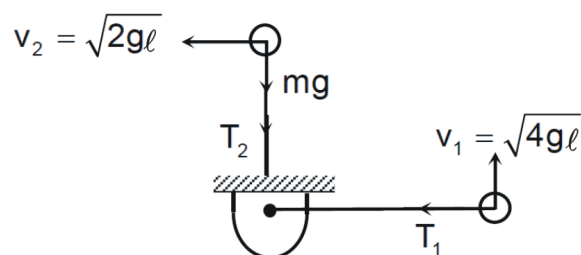
6. Ans (4)

For vertical circular motion, student should

always remember that



For given, $u = \sqrt{6gl}$



At horizontal position,

$$T_1 = \frac{mv_1^2}{l} = 4mg \quad \dots(i)$$

At highest point,

$$T_2 + mg = \frac{mv_2^2}{l} = 2mg$$

$$T_2 = mg \quad \dots(ii)$$

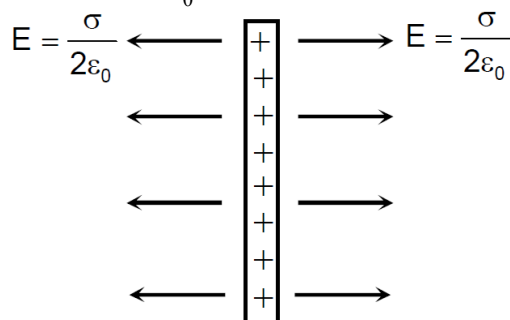
From above equation, we get

$$\frac{T_1}{T_2} = 4$$

7. Ans (4)

The electric field produced by thin charged sheet

is given by $\frac{\sigma}{2\epsilon_0}$



[Where, σ = surface charge density]

$$E_0 + E = 12 \text{ V/m} \quad \dots(i)$$

$$E_0 - E = 8 \text{ V/m} \quad \dots(ii)$$

Solving Equations (i) and (ii), we get

$$E = 2 \text{ V/m}$$

Now, electric field due to plate = $\frac{\sigma}{2E_0} = 2$

$$S = 4e_0$$

$$\therefore n = 4$$

8. Ans (6)

From the given conditions, we get

$$E_n - E_2 = (10.2 + 17) \text{ eV}$$

$$= 27.2 \text{ eV} \quad \dots(i)$$

$$E_n - E_3 = (4.25 + 5.95) \text{ eV}$$

$$= 10.2 \text{ eV} \quad \dots(ii)$$

Subtracting Equation (ii) from Eq. (i), we get

$$E_3 - E_2 = 17.0 \text{ eV}$$

$$Z^2 (13.6) \left[\frac{1}{4} - \frac{1}{9} \right] = 17.0$$

$$Z^2 (13.6) \left(\frac{5}{36} \right) = 17.0$$

$$Z^2 = 9 \text{ or } Z = 3$$

$$Z^2 (13.6) \left[\frac{1}{4} - \frac{1}{n^2} \right] = 27.2$$

$$3^2 (13.6) \left[\frac{1}{4} - \frac{1}{n^2} \right] = 27.2$$

$$\frac{1}{4} - \frac{1}{n^2} = 0.222$$

$$\Rightarrow \frac{1}{n^2} = 0.0278 \Rightarrow n^2 = 36 \Rightarrow n = 6$$

PART-2 : CHEMISTRY

SECTION-I (i)

- Ans (D)**
sp² hybridization of Hg and Hg²⁺ has d¹⁰ configuration
- Ans (A)**
MnO₂ oxidise allylic alcohols
- Ans (B)**
Factual
- Ans (A)**
Factual

PART-2 : CHEMISTRY

SECTION-I (ii)

- Ans (C,D)**
Factual
- Ans (A,C,D)**
Down the group nucleophilicity increases
- Ans (A,D)**
–NO₂ and –Br increases acidic character of phenol by dispersing –ve charge of phenoxide ion – OCH₃ and –CH₃ decreases acidic character by intensification of –ve charge of phenoxide ion.
- Ans (A,D)**
Factual
- Ans (A,C)**
p-subshell can have maximum 6 electron and an orbital can have maximum 2 electron with opposite spin.
- Ans (A,C)**
Factual

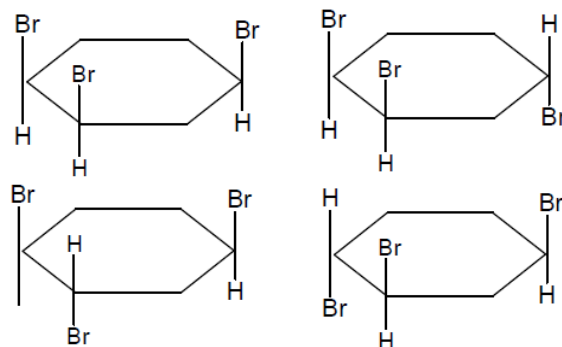
PART-2 : CHEMISTRY

SECTION-III

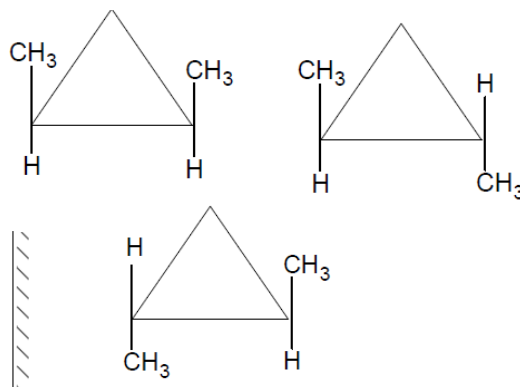
- Ans (7)**
for s-subshell, $\ell = \text{zero}$, hence for all electrons present in s-orbitals, $n \times \ell = \text{zero}$
Cr = 1s², 2s², 2p⁶, 3s², 3p⁶, 3d⁵, 4s¹
In Cr there are 7 electrons present in s-orbitals
- Ans (6)**
Factual
- Ans (6)**
Triangular bipyramidal molecules have six, 90° angles
- Ans (5)**
edta make five rings of 5 member, $x = 5$ and octahedral [M(edta)] complex can not show geometrical isomerism, $y = 0$
- Ans (6)**
 $\text{Cr}_2\text{O}_7^{2-} + 6\text{Fe}^{2+} + 14\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$

- Ans (4)**
$$Z = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$$

- Ans (4)**



- Ans (3)**

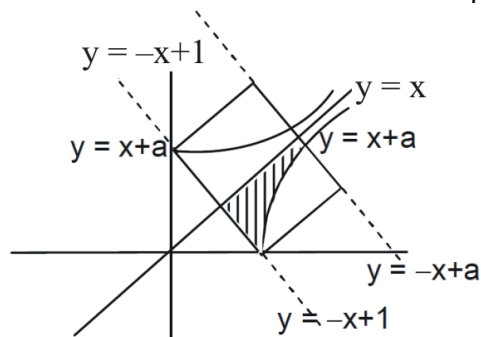


PART-3 : MATHEMATICS

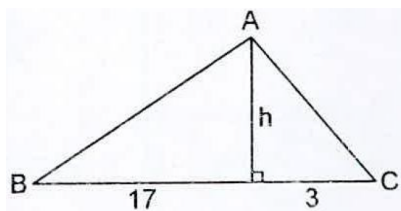
SECTION-I (i)

1. **Ans (D)**

Required area of shaded region is $k - \frac{1}{4}$



2. **Ans (B)**



$$B + C = \pi - A$$

$$\tan(B + C) = -\tan A$$

$$\frac{h\left(\frac{1}{17} + \frac{1}{3}\right)}{1 - \frac{h^2}{51}} = -\frac{22}{7} \Rightarrow \left(\frac{20h}{51 - h^2}\right) = -\frac{22}{7}$$

$$\text{Area of triangle ABC } \Delta = -\frac{1}{2} \times 20 \times 11 = 110$$

3. **Ans (B)**

Let three digits of N be :

Hundred's digit = $a - d$, ten's digit = a , and

unit's digit = $a + d$

$$(a - d) + a + (a + d) = 15 \Rightarrow a = 5$$

$$N = 100(5 - d) + 50 + (5 + d) \quad \dots (1)$$

Number obtained by reversing digits of N is

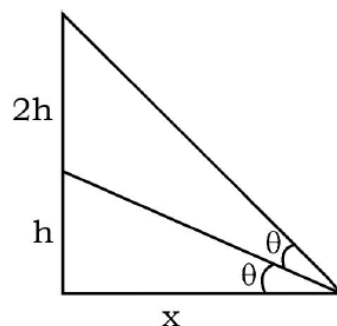
$$N_1 = 100(5 + d) + 50 + (5 - d) \quad \dots (2)$$

$$\text{Now, } 594 = N - N_1 = 10(-2d) + 2d$$

$$\Rightarrow 2d = -6 \Rightarrow d = -3$$

$$\text{Thus, } N = 852 \text{ and } \frac{1000}{N - 252} = \frac{1000}{600} = \frac{5}{3}$$

4. **Ans (C)**



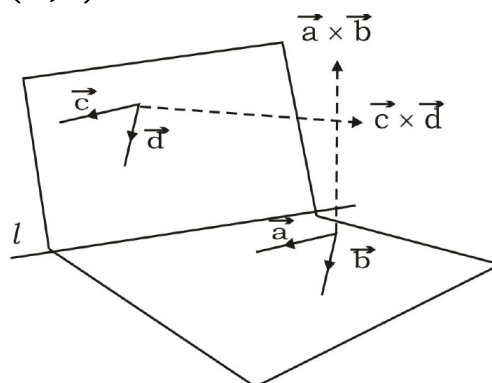
Let $\frac{h}{x} = a$. Then $\tan \theta = a$, $\tan 2\theta = 3a$.

$$\therefore \frac{2a}{1 - a^2} = 3a \Rightarrow a = \frac{1}{\sqrt{3}}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,B)**



$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$ = a vector perpendicular to $\vec{a} \times \vec{b}$ and $\vec{c} \times \vec{d}$

If l is the line of intersection of the two planes of $\vec{a}, \vec{b}$ and $\vec{c}, \vec{d}$ then $l \perp \vec{a} \times \vec{b}$ and $l \perp \vec{c} \times \vec{d}$ So,

$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$ is along l which is equally inclined to $\vec{a} \times \vec{b}$ and $\vec{c} \times \vec{d}$

6. **Ans (A,B,C,D)**

$$(A) {}^7C_2 - {}^3C_2 - {}^5C_2 + 2 = 10$$

$$(B) {}^7C_3 - {}^3C_3 - {}^5C_3 = 24$$

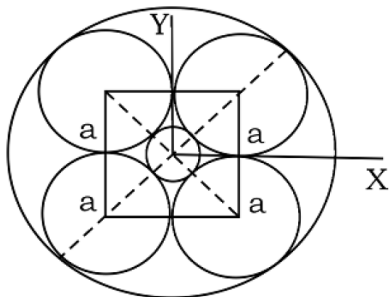
(C) There is no use of point '3'

$${}^6C_4 - {}^4C_4 - {}^4C_3 \cdot {}^2C_2 = 6$$

$$(D) {}^7C_3 - {}^3C_3 - {}^5C_3 = 24$$

7. **Ans (A,C)**

The four circles are as shown in the figure. The smallest circle touching all of them has the radius $= \sqrt{2}a - a$ and the greatest circle touching all of them has the radius $= \sqrt{2}a + a$.



8. **Ans (A,C)**

Consider $f(x) = x^3 + ax^2 + bx - 2$

$$\alpha \beta \gamma = 2$$

$a, b, \gamma \in \mathbb{N} \Rightarrow$ two of them are 1 & other is 2

$$\Rightarrow a = -(\alpha + \beta + \gamma) = -4 = \frac{f''(0)}{2}$$

$$b = (\alpha\beta + \beta\gamma + \alpha\gamma) = 5 = f'(0)$$

$$\text{Hence } f(x) = x^3 - 4x^2 + 5x - 2$$

Again $\because$ 1 is a repeated root of $f(x) = 0$

$$\Rightarrow f'(1) = 0$$

9. **Ans (A,B,C,D)**

$$x + [y] + \{z\} = 1.1 \quad \dots(1)$$

$$[x] + \{y\} + z = 2.2 \quad \dots(2)$$

$$\{x\} + y + [z] = 3.3 \quad \dots(3)$$

$$(1) + (2) + (3)$$

$$\Rightarrow 2(x + y + z) = 6.6$$

$$\Rightarrow x + y + z = 3.3 \quad \dots(4)$$

$$(4) - (1)$$

$$\{y\} + [z] = 2.2 \Rightarrow \{y\} = 0.2 \text{ \& } [z] = 2$$

$$(4) - (2)$$

$$\{x\} + [y] = 1.1 \Rightarrow \{x\} = 0.1, [y] = 1$$

$$(4) - (3)$$

$$[x] + \{z\} = 0 \Rightarrow [x] = 0 \text{ \& } \{z\} = 0$$

$$\begin{cases} x = 0.1 \\ y = 1.2 \\ z = 2 \end{cases}$$

10. **Ans (B,D)**

$$b^2 = ac$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{a^2 + c^2 - b^2}{2b^3}$$

$$\therefore \cos B = \frac{a^2 + c^2}{2ac} - \frac{1}{2}$$

$$\Rightarrow \cos B + \frac{1}{2} = \frac{a^2 + c^2}{2ac} \Rightarrow \cos B + \frac{1}{2} \geq 1 \Rightarrow \cos B \geq \frac{1}{2}$$

$$B \in (0, \pi/3)$$

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (9)**

$$S = 3.3^{2/2} . 3^{3/4} . 3^{4/8} \dots \dots \infty$$

$$S = 3^{1 + \frac{2}{2} + \frac{3}{4} + \frac{4}{8} + \dots \dots \infty}$$

$$S' = 1 + \frac{2}{2} + \frac{3}{4} + \frac{4}{8} + \dots \dots \infty \quad \dots(1)$$

$$\frac{S'}{2} = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \dots \dots \infty \quad \dots(2)$$

Subtract (2) from (1)

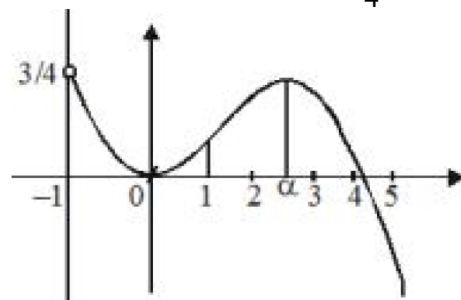
$$\frac{S'}{2} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \dots \infty$$

$$S' =$$

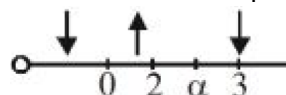
$$S = 3^4 = 81$$

2. **Ans (2)**

$$f(x) = (1+x)\ln(1+x) - x - \frac{x^2}{4}$$

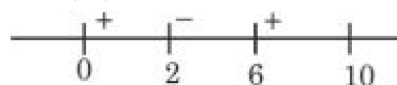


$$\lim_{x \rightarrow -1} f(x) = 0 + 1 - \frac{1}{4} = \frac{3}{4} \text{ and } f(0) = 0$$



$$\Rightarrow \alpha \in (2, 3)$$

3. **Ans (2)**



4. Ans (9)

Projection length = $|\vec{a} \cdot \vec{p}|$

$$\text{So, } \frac{|x + 2y + 3z|}{\sqrt{14}} = 6$$

$$\Rightarrow |x + 2y + 3z| = 6\sqrt{14}$$

$$\Rightarrow \left| (x\hat{i} + \sqrt{2}y\hat{j} + \sqrt{3}z\hat{k}) \cdot (\hat{i} + \sqrt{2}\hat{j} + \sqrt{3}\hat{k}) \right| = 6\sqrt{14}$$

$$\Rightarrow (x^2 + 2y^2 + 3z^2) (1 + 2 + 3) \cos^2 \theta = (6\sqrt{14})^2$$

$$\Rightarrow \frac{x^2 + 2y^2 + 3z^2}{\sqrt{14}} \geq 6\sqrt{14} \Rightarrow 1 = 6\sqrt{14}$$

$$\text{So, } (t^2 + 15^2)^{1/3} = (504 + 225)^{1/3} = (729)^{1/3} = 9$$

5. Ans (0)

$$\vec{c} = \frac{\vec{a} - \mu\vec{b}}{4} \text{ and } \vec{a} \cdot \vec{b} = 0$$

Now,

$$\begin{aligned} (\vec{b} - \vec{a}) \cdot (\vec{b} + \vec{c}) &= 0 \Rightarrow (\vec{b} - \vec{a}) \cdot \left(\vec{b} + \frac{\vec{a} - \mu\vec{b}}{4} \right) = 0 \\ \Rightarrow (4 - \mu) b^2 &= a^2 (\because \mu < 4) \quad \dots(i) \end{aligned}$$

Again

$$\begin{aligned} 4|\vec{b} + \vec{c}|^2 &= |\vec{b} - \vec{a}|^2 \Rightarrow 4 \left| \frac{(4 - \mu)\vec{b} + \vec{a}}{4} \right|^2 = |\vec{b} - \vec{a}|^2 \\ \Rightarrow 4 \left(\frac{4 - \mu}{4} \right)^2 b^2 + \frac{a^2}{4} &= b^2 + a^2 \Rightarrow ((4 - \mu)^2 - 4) b^2 = 3a^2 \\ \dots(ii) \end{aligned}$$

(i) & (ii) we get

$$\begin{aligned} \frac{(4 - \mu)^2 - 4}{4 - \mu} &= \frac{3}{1} \Rightarrow \mu^2 - 5\mu = 0 \\ \Rightarrow \mu &= 0 \text{ or } 5 \text{ but as } \mu < 4, \text{ so, } \mu = 0. \end{aligned}$$

6. Ans (8)

$$\begin{aligned} p &= \frac{{}^{5!-3}C_1 (4!) \cdot (2!) + {}^3C_2 (3!) \cdot (2!) \cdot (2!) - {}^3C_2 (2!) \cdot (2!) \cdot (2!) \cdot (2!)}{(5!)} \\ &= \frac{120 - 144 + 72 - 16}{120} = \frac{4}{15} \\ \text{So } 30p &= 8 \end{aligned}$$

7. Ans (2)

$$\int_I x f''(2x) dx = x \frac{f'(2x)}{2} - \int \frac{f'(2x)}{2} dx = x \frac{f'(2x)}{2} - \frac{f(2x)}{4}$$

$$\int_0^1 x f''(2x) dx = \left[\frac{1}{2} x f'(2x) - \frac{f(2x)}{4} \right]_0^1 = \left(\frac{f'(2)}{2} - \frac{f(2)}{4} \right) - \left(0 - \frac{f(0)}{4} \right)$$

$$\left(\frac{5}{2} - \frac{3}{4} \right) - \left(0 - \frac{1}{4} \right) = 2$$

8. Ans (1)

Let E_1, E_2, E_3, E_4, E_5 and E_6 be the events of

occurrence of 1, 2, 3, 4, 5 and 6 on the

dice respectively, and let E be the event

$$\therefore P(E_1) = \frac{1-k}{6}; P(E_2) = \frac{1+2k}{6}; P(E_3) = \frac{1-k}{6}$$

$$P(E_4) = \frac{1+k}{6}; P(E_5) = \frac{1-2k}{6}; P(E_6) = \frac{1+k}{6}$$

$$\text{and } \frac{1}{9} \leq P(E) \leq \frac{2}{9}$$

Then, $E \equiv \{(3, 6), (6, 3), (4, 5), (5, 4)\}$

Hence, $P(E) = P(E_3 E_6) + P(E_6 E_3) + P(E_4 E_5)$

+ $P(E_5 E_4)$

$$= P(E_3) P(E_6) + P(E_6) P(E_3) + P(E_4) P(E_5)$$

$$+ P(E_5) P(E_4)$$

$$= 2P(E_3) P(E_6) + 2P(E_4) P(E_5)$$

{Since E_1, E_2, E_3, E_4, E_5 and E_6 are

independent}

$$= 2 \left(\frac{1-k}{6} \right) \left(\frac{1+k}{6} \right) + 2 \left(\frac{1+k}{6} \right) \left(\frac{1-2k}{6} \right)$$

$$= \frac{1}{18} [2 - k - 3k^2]$$

$$\text{Since, } \frac{1}{9} \leq P(E) \leq \frac{2}{9}$$

$$\therefore -\frac{1}{3} \leq k \leq 0$$

$\therefore$ Set of integral value of $k = \{0\}$

$\therefore$ Number of integral solution of k is 1